FNH 200 942

Lesson 11 - Nutrient Retention

Also, a sample question on the final exam

2026 Summer

Team Project

Please add t
the wiki:

https://wiki.ubc.ca/Course:FNH200/Projects/2026

summary here, and contribute your final exam question

2. Modify an existing page created by past FNH 200 students: Past Projects
3. Create a page on an endangered food, or a food with cultural and historic significance, AND not yet explored by any past FNH 200 students

Contents hide

Team Number	Product Investigated and Link	Completed? To be updated by Aug 13
0 (Example)	The Processing and Safety of Swiss Cheese (Please provide link to the page generated above)	No
1	• Course:FNH200/Projects/2026/Nescafe Coffee Pods/Capsules - What is inside?	No
2	• Bear root (endangered food): Course:FNH200/Projects/2026/Bear root (endangered food)	No
3	• Ice Cider (endangered food): Course:FNH200/Projects/2026/Ice Cider - Exploring the Endangered Drink	No
4	Course:FNH200/Projects/2023/Olestra Regulation	No
5	• If you are enhancing an existing page, add that existing link here • If you are doing an interview or researching an endangered food, create a new link using the Create Your Project Page tool below and add the link here	No
6	Pu-erh Tea (Endangered Food) Course:FNH200/Projects/2026/Pu-erh	No

Final Peer Review

Course Overview:

There are 13 lessons in FNH 200, in varying lengths.

Foundation

- **Lesson 01** Food Science and Canadian Food System
- **Lesson 02** Chemical and Physical Properties of Foods
- **Lesson 03** Fat and Sugar Substitutes
- **Lesson 04** Food Standards, Regulations, and Guides

Preservation

Lesson 05

- **Lesson 06** Thermal
- **Lesson 07** Low Temperature
- **Lesson 08** Dehydration
- **Lesson 09** Biotechnology
- **Lesson 10** Irradiation

Healthy & Safe Future

- **Lesson 11** Affects on Nutrient Retention
- **Lesson 12** Toxicants in Foods and Foodborne Diseases
- **Lesson 13** Trends in Food for Nutrition and Health

Learning Objectives

After reading the article by Newsome, you should be able to:

1. Make a list of benefits versus risks (or "advantages and disadvantages") of food processing.
2. Compare and contrast the stability of the following groups of nutrients:
   1. Vitamins A, C, and E
   2. Thiamin (B1), niacin (B2), and pyridoxine (B6)
3. Identify main factors that affect the extent of nutrient loss as a result of heat processing, based on tables 3, 4, and 5
4. Identify the one preservation method that results in a nutritive quality closest to that of fresh, raw foods
5. Name the process that is also referred to as 'cold sterilization'
6. List at least two ways to minimize nutrient loss during storage
7. List at least five strategies to minimize nutrient loss as we prepare foods at home
8. Compare and contrast the following terms: restoration, fortification and enrichment

Required Reading

Download from Canvas:

Newsome, R.L. 1986 [December]. (A scientific status summary). Food Technology, 40(12):109-116.

A classic!

OFFICE OF SCIENTIFIC PUBLIC AFFAIRS **SCIENTIFIC STATUS SUMMARY**

A PUBLICATION OF THE INSTITUTE OF FOOD TECHNOLOGISTS

Effects of Food Processing on Nutritive Values

Scientific Status Summary by the Institute of Food Technologists' Expert Panel On Food Safety & Nutrition

Food processing began thousands of years ago when early man preserved his harvest, fresh meats, and fish for winter reserves by drying in the sun, later expanding this to smoking and salting these foods. Our forefathers labored continually, because seasonal availability and climate necessitated strict attention to schedules. Fruits and vegetables needed to be picked at the right time, animals had to be slaughtered when and where they could be found and

i.e., flavor, odor, texture, and appearance. Home processing results in a more variable product than commercial processing, and nutrient losses are frequently higher. However, processing is essential, since the concentrations of various nutrients decrease in unprocessed foods and eventually wholesomeness is lost.

Some losses in certain nutrients during food processing are inevitable. The major consideration in evaluating food processing from a

Processing Losses Versus Natural Differences

Variations in the nutrient content of raw food materials may affect the content of vitamins and minerals in the final product as much as, and sometimes more than, processing itself. Raw foods may vary widely in their vitamin content as a result of genetic differences, climatic or soil condi-

XL Foods dumps tonnes of recalled meat at landfill

Disposal comes after widespread recall of meat products possibly contaminated with E. coli

CBC News Posted: Oct 21, 2012 4:38 PM MT | Last Updated: Oct 21, 2012 5:57 PM MT

What do you know about nutrient loss during processing?

Which nutrient below is MOST stable under thermal processing?

A. Carbohydrates
B. Protein
C. Fat
D. Vitamins
E. Minerals

Processing increases bioavailability of nutrients by destroying:

A. Amylase inhibitors
B. Thiaminase
C. Avidin
D. All of the above
E. None of the above

Bottom of p. 109

Top of p. 110

the thiamin intake of the animal.

Processing Effects: Pluses and Minuses

The basic food preservation methods used by early man are still utilized today. These and other newer methods are listed in Table 1. An analysis of these processing techniques reveals both favorable and unfavorable effects on nutritional quality.

On the positive side, heat processing destroys antidigestive factors such as trypsin and amylase inhibitors in cereal grains, peas, and beans, thus improving the bioavailability (digestibility) of the proteins and carbohydrates in these products. Heat processing also destroys thiaminase, which destroys thiamin in fish, shellfish,

brussels sprouts, and red cabbage, and it destroys the avidin and other factors in raw egg white that would otherwise bind biotin and some of the iron present in egg yolk and make these nutrients biologically unavailable. Heat processing increases the digestibility of starch and protein (by gelatinization and denaturation, respectively), and it increases the bioavailability of niacin, which is present in many cereals in a bound form. Heat processing also increases the palatability of food, resulting in increased appeal and nutrient consumption.

Which vitamin below is LEAST stable in normal food pH (<7)?

A. Vitamin A
B. Vitamin C
C. Vitamin E

Table 2—Stability of Nutrients[a,b]

	Effect of food pH			Effect of food environment		
Nutrient	Neutral (pH 7)	Acid (<pH 7)	Alkaline (>pH 7)	Air or oxygen	Light	Heat
Vitamin A	S	U	S	U	U	U
Vitamin C	U	S	U	U	U	U
Biotin	S	S	S	S	S	U
Beta-carotene	S	U	S	U	U	U
Choline	S	S	S	U	S	S
Cobalamin (B-12)	S	S	S	U	U	S
Vitamin D	S	S	U	U	U	U
Folic acid	U	U	S	S	U	S
Inositol	S	S	S	S	S	U
Vitamin K	S	U	U	S	U	S
Niacin	S	S	S	S	S	S
Pantothenic acid	S	U	U	S	S	U
Pyridoxine (B-6)	S	S	S	S	S	S
Riboflavin	S	S	U	S	U	U
Thiamin	U	S	U	U	S	U
Vitamin E	S	S	S	U	U	U
Mineral salts	S	S	S	S	S	S

[a]Modified from Harris (1975)
[b]S = stable (no important destruction); U = unstable (significant destruction)

Which processing below will lead to most loss of Vitamin C? Tables 3, 4, and 5

A. Blanching in water
B. Blanching with steam
C. Commercial sterilization
D. High temperature, short time process
E. Ultra high temperature process

Table 3—Effects of Blanching on Nutrients in Vegetables[a,b]

Blanching method	Nutrient	% Loss
Water	Vitamin C	16–58
	Riboflavin	30–50
	Thiamin	16–34
	Niacin	32–37
Steam	Vitamin C	16–26
	Vitamin B-6	21

[a]Although other blanching systems are available (e. g., hot gas, microwave, and individual quick blanching), data on nutrient retention were not sufficiently complete to allow reporting percentage loss
[b]From Lund (1982)

Table 4—Effects of Conventional Canning on Nutrients in Vegetables[a]

Nutrient	% Loss
Vitamin C	33–90
Thiamin	16–83
Riboflavin	25–67
Niacin	0–75
Folacin	35–84
Pantothenic acid	30–85
Vitamin B-6	0–91
Biotin	0–78
Vitamin A	0–84

[a]From Lund (1982)

Table 5—Effect of Aseptic (HTST[a]) and Conventional Thermal Processing Methods on Nutrient Losses[b]

	Thiamin loss (%)		Pyridoxine loss (%)	
Product	HTST	Conventional	HTST	Conventional
Strained lima beans	15.8	40.3	9.5	10.1
Strained beef	9.5	21.6	4.1	2.9
Tomato juice concentrate	0	2.8	0	0

[a]From Everson et al. (1964a; b)
[b]High-temperature/short-time aseptic processing

Which processing below leads to the LEAST nutrient loss?

A. Pasteurization
B. Ultra high temperature processing
C. Refrigeration
D. Freezing
E. Irradiation

Freezing. Freezing results in a nutritive quality closest to that of fresh, raw foods. In general, nutrient loss during freezing is negligible using proper packaging (air impermeable) and processing conditions. Exceptions are small losses of vitamin C and other water- soluble vitamins in vegetables and fruits during blanching and small (10–20%) unexplainable losses of water-soluble vitamins in pork. Proper freezing conditions are important to nutrient retention. Foods which are not frozen quickly, and solidly, and maintained at a constant low temperature (−18°F or lower), or which undergo intermittent thawing, exhibit a greater than normal drip loss when thawed and, hence, greater nutrient loss. The shelf life of frozen foods (9 months to 1 year) is shorter than that of canned foods because not all of the water in food freezes and because some fatty acids as well as vitamins A, C, and E tend to oxidize during storage when exposed to air. This allows some chemical changes to occur, even in the frozen state.

Name the process that is also known as 'Cold Sterilization'. Middle of p. 113

A. Commercial sterilization
B. Individual quick frozen
C. Freeze drying
D. Fermentation at low temperature
E. Irradiation

Irradiation. The irradiation process has been used for many years in countries other than the U.S., and in the U.S. to disinfect dry or dehydrated spices and herbs and to sterilize food for astronauts and for hospital patients with immune system disorders. Radiation energy absorbed by food during irradiation is expressed in "rads" or "Grays" — one Gray (Gy) = 100 rads; one kiloGray (kGy) = 100,000 rads. Low-dose irradiation (up to 1 kGy) was recently approved in the U.S. (April 1986) for use in fruits and vegetables to control maturation and insects, and for use in fresh pork (minimum of 0.3 kGy) to reduce the risk of trichinosis (July 1986). Low-dose irradiation may also be used to inhibit sprout development in white potatoes (50–150 Gy) and to control insects in wheat and wheat flour (200–500 Gy) (approved in August 1964 and August 1963, respectively). Use of irradiation for these purposes, however, is much more common in other countries.

Which techniques below should NOT be used to minimize nutrients loss during storage? pp. 113 and 114

A. Storage at low temperature
B. Storage at high temperature
C. Addition of antioxidants
D. Use of protective coatings
E. Vacuum packaging
F. Packaging in tin foil

Storage. Nutrient losses can occur during transportation and storage of fresh foods. Foods which are processed with good manufacturing practices from high-quality, freshly harvested, "garden-fresh" commodities will frequently have a higher nutrient content than freshly harvested "market-fresh" commodities which have been improperly handled during trans-

portation and/or which have been stored for a few days or more.

Vitamin losses during storage and distribution of canned or dried foods vary widely, depending on the temperatures to which they are exposed during distribution and storage. Retention of a number of vitamins in canned tomato juice, for example, is markedly decreased by storage at temperatures higher than room temperature. However, storage of canned goods for up to one year at temperatures lower than 70–75°F results in less than 10–20% change in the concentration of vitamins. The storage stability of vitamins and minerals added to breakfast cereals (Table 6) is excellent.

Oxidation and browning may occur in some dried foods during storage. Oxidation of fatty acids results in rancidity and an inedible product. Nonenzymatic browning (Maillard reaction) occurs between reducing sugars such as fructose and the epsilon amino group of the essential amino acid lysine, decreasing the bioavailability of this amino acid. This reaction may occur in foods processed by other methods as well, since it is

8. According to the article, this food has been...

A. Restored

Adding back what's lost

B. Fortified

Adding something that's not naturally in the food

C. Enriched

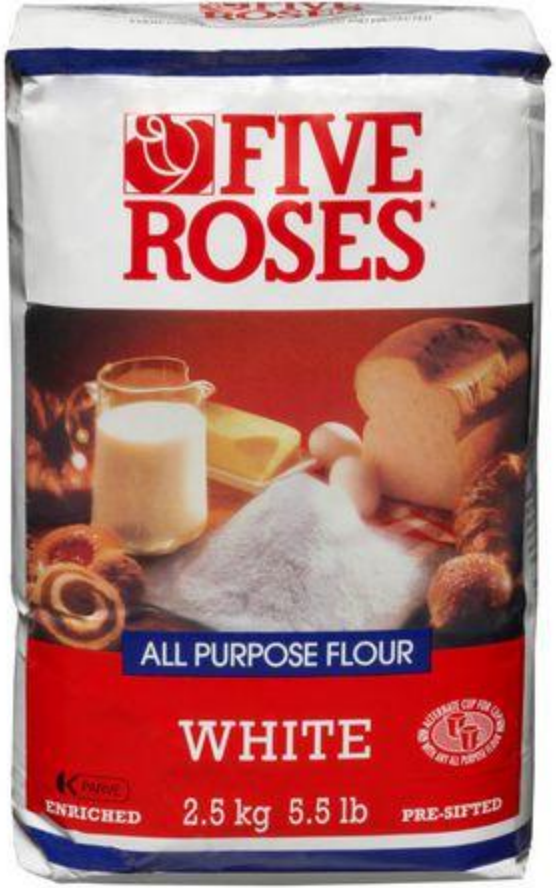
Similar to fortification, but to meet some standards, and usually more than naturally presented

9. This food has been...

A. Restored
B. Fortified
C. Enriched

10. This food has been...

A. Restored
B. Fortified
C. Enriched



Fortifications

The next three slides are included here as students in the past were keen to learn more about fortifications of foods in Canada.

They will not be included in the final exam.

In Canada, Mandatory Vs Voluntary Fortification

Foods to Which Vitamins, Mineral Nutrients and Amino Acids May or Must be Added

Column 1 Food	Column 2 Vitamin, Mineral Nutrient or Amino Acid
1. Breakfast cereals	**Voluntary:** Thiamine, niacin, vitamin B6, folic acid, pantothenic acid, magnesium, iron, zinc
2. Fruit nectars, vegetable drinks, bases and mixes for vegetable drinks and a mixture of vegetable juices	**Voluntary:** Vitamin C
2.1 Fruit flavoured drinks that meet all the requirements of B.11.150, FDR	**Mandatory:** Vitamin C **Voluntary:** Folic acid, thiamine, iron, potassium

Breakfast Cereal

B.13.060 Notwithstanding sections D.01.009, D.01.011 and D.02.009, no person shall sell a breakfast cereal to which a vitamin or mineral nutrient set out in Column I of an item of the table to this section has been added, either singly or in any combination, unless each 100 g of the breakfast cereal contains the added vitamin or mineral nutrient in the amount set out in Column II of that item.

TABLE

	Column I	Column II
Item	Vitamin or Mineral Nutrient	Amount per 100 g of Breakfast Cereal
1	Thiamine	2.0 mg
2	Niacin	4.8 mg
3	Vitamin B_6	0.6 mg
4	Folic Acid	0.06 mg
5	Pantothenic Acid	1.6 mg
6	Magnesium	160.0 mg
7	Iron	13.3 mg
8	Zinc	3.5 mg

B.08.003 [S]. Milk or **Whole Milk**

(a) shall be the normal lacteal secretion obtained from the mammary gland of the cow, genus *Bos*; and

(b) shall contain added vitamin D in such an amount that a reasonable daily intake of the milk contains not less than 300 International Units and not more than 400 International Units of vitamin D.

SOR/95-499, s. 2.

B.08.004 [S]. Skim Milk

(a) shall be milk that contains not more than 0.3 per cent milk fat;

(b) shall, notwithstanding sections D.01.009 and D.01.010, contain added vitamin A in such an amount that a reasonable daily intake of the milk contains not less than 1,200 International Units and not more than 2,500 International Units of vitamin A; and

(c) shall contain vitamin D in such an amount that a reasonable daily intake of the milk contains not less than 300 International Units and not more than 400 International Units of vitamin D.

SOR/78-656, s. 1.

Sample Final Exam Question - Lasagna

Lasagna - How does cooking affect its nutritive value?

Spinach:	Vitamin C
Tomato:	Vitamin A
Beef:	Fat, Protein
Cheese:	Calcium
Pasta:	Carbohydrate
	Thiamine

	Heating at pH>4.6 (make an assumption)	Effect?
Spinach, Vitamin C	Stable, Unstable????	Lost by a bit, a lot???
Tomato, Vitamin A	Stable, Unstable????	
Beef, Fat	Impact of cooking on fat	
Beef, Protein		
Cheese, Calcium		
Pasta, Carbohydrate	Digestive inhibitor	bioavailability
Pasta, Thiamine	Refer to table 2	

Lasagna - How does freezing affect its nutritive value?

Spinach: Vitamin C

Tomato: Vitamin A

Beef: Fat, Protein

Cheese: Calcium

Pasta: Carbohydrate

Thiamine